LAB 112

UICT · Professional Certificate in Software Engineering

Real-Life Project II

Project Proposal Ideas

Solving Real Problems in Uganda with Software Engineering

2025 · Kampala, Uganda

About This Document

This document contains 8 project proposal ideas designed specifically for UICT PCSE students undertaking Real-Life Project II. Each project targets a real problem in Uganda or the local community and proposes a software engineering solution that is achievable with the skills you have covered so far — HTML, CSS, JavaScript, SQL, and the C# and Java concepts currently being taught.

Each proposal includes: a clear problem description, the target audience, core features, a technology stack, and expected outcomes. Use these as inspiration for your own proposal — you can take one of these ideas and adapt it, or let it spark an entirely new idea of your own.

Note: A good project proposal focuses on a specific, real problem. The simpler and more focused your problem, the stronger your proposal will be.

Project 01 [Web Application]

School Fee Tracker

Target Audience	Parents and school administrators in Ugandan primary and secondary schools
Problem	Many parents in Uganda struggle to keep track of school fee balances, payment deadlines, and outstanding arrears. Schools also spend a lot of time manually recording payments in books, leading to errors and lost records. There is no easy way for a parent to check how much is owed without physically going to school.
Core Features	• Parent dashboard showing balance, amount paid, and amount remaining • Payment history with dates and amounts • Deadline reminders for upcoming fee instalments • Admin panel for school to record and update payments • Student profile per class with fee status
Tech Stack	HTML, CSS, JavaScript (frontend) · SQL database for storing payment records · C# for backend logic
Expected Outcome	Reduces manual record keeping for schools, gives parents clear visibility of fee balances, and reduces misunderstandings between parents and administration.

Project 02 [Web Application]

Campus Lost & Found Portal

Target Audience	Students and staff at UICT and other Ugandan campuses
Problem	Students at UICT frequently lose items like phones, ID cards, books, and bags on campus. Currently there is no organised system to report or search for lost items — people just post on WhatsApp groups which gets cluttered and ignored quickly.
Core Features	• Submit a lost item report with description, location found, and photo • Search lost items by category or keyword • Claim an item by providing identifying details • Admin verifies and marks items as returned • Notification when a matching item is reported
Tech Stack	HTML, CSS, JavaScript (frontend) · SQL database for item records · Basic form handling
Expected Outcome	Centralises lost and found reporting on campus, increases chances of students recovering their items, and reduces cluttered social media posts.

Project 03 [Web Application]

Boda Boda Booking System

Target Audience	Commuters and boda boda riders in Kampala and surrounding areas
Problem	Boda boda transport is one of the most common ways people move around Uganda, but booking is done informally — you stand by the road and wave one down. There is no way to know the price upfront, no safety record of who you rode with, and no review system for riders.
Core Features	• Rider registration with name, photo, and contact • Customer can book a ride by entering pickup and destination • Estimated fare shown before booking • Ride history for both rider and customer • Star rating and review system after each ride
Tech Stack	HTML, CSS, JavaScript (frontend) · SQL for rider and booking records · C# backend logic
Expected Outcome	Makes boda boda transport safer and more organised, builds trust between riders and customers, and creates a digital record of rides.

Project 04 [Web Application]

Community Health Clinic Queue System

Target Audience	Patients and staff at local government health centres in Uganda
Problem	Government health centres in Uganda like Kiswa, Kawaala, and others are overwhelmed with patients who have to queue for hours, sometimes from early morning. There is no system to manage patient flow — people crowd at the door not knowing when they will be seen.
Core Features	• Patient registers on arrival and gets a queue number • Display board showing current number being served • Estimated wait time shown to patient • Nurse/doctor marks each patient as done to advance the queue • Daily patient count report for the clinic
Tech Stack	HTML, CSS, JavaScript (frontend) · SQL for patient queue records · Simple backend logic
Expected Outcome	Reduces chaos at health centres, improves patient experience, gives staff visibility of how many patients are waiting, and allows better resource planning.

Project 05 [Web Application]

Student Study Group Finder

Target Audience	University and college students across Uganda, starting with UICT
Problem	Many students at UICT study alone even when they are struggling, simply because they don't know which of their classmates are available to study or are working on the same topics. There is no platform to coordinate study groups beyond informal WhatsApp messages.
Core Features	• Student creates a profile with their course and subjects • Post a study session with topic, date, time, and location • Browse and join available study groups • View who has joined your group • Mark a session as completed
Tech Stack	HTML, CSS, JavaScript (frontend) · SQL for student and session records
Expected Outcome	Encourages peer learning, reduces student isolation, and makes it easier to organise focused study sessions around specific topics or assignments.

Project 06 [Web Application]

Local Market Price Checker

Target Audience	Shoppers, vendors, and households in Kampala markets like Owino, Nakasero, and Kalerwe
Problem	Prices for everyday goods like food, vegetables, and household items vary greatly between markets and even between stalls in the same market. A buyer visiting Owino market has no idea if the price they are being quoted is fair or inflated compared to other stalls or other markets.
Core Features	• Browse current prices for common goods by market • Vendors can update prices for their stall • Price comparison between two markets for the same item • Price trend over time — is a product getting cheaper or expensive? • Search for a specific item across all listed markets
Tech Stack	HTML, CSS, JavaScript (frontend) · SQL for price and market records
Expected Outcome	Empowers buyers with price information, encourages fair pricing by vendors, and helps households budget more accurately for daily shopping.

Project 07 [Web Application]

Church / Mosque Event & Announcement Board

Target Audience	Religious congregation members and administrators in Uganda

Problem	Large churches and mosques in Uganda like Namirembe Cathedral, Rubaga, or Old Kampala Mosque have thousands of members. Announcements are currently made verbally during services or via printed flyers — members who miss a service miss all announcements. WhatsApp groups become unmanageable at large scale.
Core Features	• Post upcoming events with date, time, venue, and description • Pinned important announcements at the top • Archive of past events and announcements • Members can RSVP to events • Admin approves before any post goes public
Tech Stack	HTML, CSS, JavaScript (frontend) · SQL for events and members
Expected Outcome	Keeps congregation members informed and engaged, reduces missed announcements, and gives the organisation a professional online presence.

Project 08 [Web Application]

Second-Hand Textbook Exchange

Target Audience	University and college students across Uganda
Problem	Textbooks in Uganda are expensive. Many students buy books new, use them for one semester, and then have no one to sell them to. The next intake of students then buys the same books new at full price. There is no organised platform for students to buy and sell used textbooks within their institution.
Core Features	• List a textbook for sale with photo, title, author, condition, and price • Browse available textbooks by course or subject • Contact seller directly through the platform • Mark a book as sold once exchanged • Wishlist — post that you are looking for a specific book
Tech Stack	HTML, CSS, JavaScript (frontend) · SQL for listings and users
Expected Outcome	Saves students money on textbooks, reduces waste of unused books, and creates a sustainable academic resource sharing culture on campus.

A Note on Your Proposal

When writing your own proposal, structure it around these five areas:

1. Problem Statement: Describe the specific problem clearly. Who experiences it? How does it affect them? Use real examples from your community.

2. Proposed Solution: Describe what your software will do to solve this problem. Keep it simple and specific.

3. Target Audience: Who will use your application? Be as specific as possible — 'students at UICT' is better than just 'students'.

4. Core Features: List 4–6 key features your application will have. Don't over-promise — focus on what you can actually build.

5. Technology Stack: List the tools and languages you will use. Be honest about what you know — HTML, CSS, JavaScript, SQL are perfectly acceptable for this level.